

to be added back into the estimated cost in order to provide investors with the full portfolio trading cost estimate.

Additionally, to simplify calculations going forward, we exclude the price appreciation term from the cost function. We only include market impact cost.

PORTFOLIO OPTIMIZATION TECHNIQUES

Quadratic Programming Approach

The trader's dilemma can be solved using a quadratic optimization (QP) by making a couple of changes to the formulation. First, formulate the problem in terms of the residual trade schedule. Second, we use the variance expression for risk, which does not include the square root function, instead of the standard deviation expression of risk, which includes the square root expression. Third, we use a variance aversion parameter in place of the traditional risk aversion parameter.

This will allow us to now solve the problem via a traditional quadratic optimization. The only outstanding issue, however, is determining the exact solution at the investor's specified level of risk aversion. This can be solved as follows. Recall that the risk aversion parameter is equal to the negative tangent of the efficient trading frontier (ETF) at the optimal trading strategy. If we solve sets of our QP optimization and plot the ETF, i.e., market impact as a function of risk using the square root function for all optimization results, we can determine the strategy on the ETF where the slope of the tangent is equal to the negative of the investor's risk aversion. This may take several iterations but it is entirely feasible.

This is an entirely valid transformation since cost-variance can be mapped to cost-risk and is consistent with Markowitz (1952) mean-variance optimization. Markowitz actually presented an optimization using return and variance but then plotted the trade-off using return and standard deviation. Markowitz's efficient investment frontier shows the trade-off between return and standard deviation but is solved using return and variance. The biggest difference here is that traders are seeking an exact point on the frontier and in an amount of time that will be useful for trading.

The QP trade cost minimization is written in terms of the residual schedule as follows:

$$\text{Min}_r \sum_{i=1}^m \sum_{t=1}^n \frac{b_1 \cdot I_i^* \cdot (r_{it} - r_{it+1})^2}{X_i \cdot v_{it}} + \lambda^* \cdot \sum_{k=1}^n r'_k C r_k \quad (9.6)$$